UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa

FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY
DEPARTMENT OF COMPUTING AND TECHNOLOGY
ADVENT 2026 SEMESTER ASSIGNMENT ONE
(25 Marks)

PROGRAM: MASTER OF SCIENCE IN DATA SCIENCE

COURSE NAME: DATA SCIENCE LIFECYCLE

The objective of this assignment is to apply exploratory data analysis and statistical techniques to the automobile dataset in order to identify the key variables that influence car price.

You have been provided with the automobileEDA.csv dataset. Use it to find main variables/features that have the most impact on the car price. **Questions:**

1. Explore the features in the dataset that could be used to predict the price of a car. Check for missing values, duplicates and outliers, and handle any you find. Save the cleaned dataset as YourLastName_Data.csv. **[5 Marks]**

2. Analyse patterns and run descriptive statistics on the dataset. Give summary statistics for the continuous variables and frequency counts for the categorical variables. Use graphs to show how price and the other variables are distributed, and summarize the findings on the patterns observed. **[5 Marks]**

3. Group the data by identified parameters, such as drive-wheels and body-style. Find the average price for each group and present the results using visuals and statistics. Which groups have the highest and lowest prices? **[5 Marks]**

4. Identify the effect of the independent attributes on price using visuals and statistics.

   a) Continuous variables vs. price **[2 Marks]**
   b) Categorical variables vs. price **[2 Marks]**
   c) Two categorical variables **[1 Marks]**

5. Based on your findings, identify the main variables that have the most impact on car price, and support your answer with a feature selection method. Write a short recommendation for a car dealer setting prices. **[5 Marks]**

Submission instructions:
Submit a single zipped file named using the format Assignment1_YourLastName. This should have the cleaned dataset as well as executed notebook. Also, Submit a link to your GitHub or Kaggle account showing your Jupyter notebook. It must include your Python code and an explanation of every output.

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